Exercise 23.1

Pick a variable that is supposedly bell-shaped in probability distribution, gather data, and make a plot of the variable's empirical distribution. Show the distribution as a histogram on a log scale and investigate whether the tails are well-modelled by a Gaussian distribution. [One example of a variable to study is the amplitude of an audio signal.]

Solution:

Exercise 23.2

Imagine that we reparameterize a positive variable x in terms of its cube root, $u = x^{1/3}$. If the probability density of x is the improper distribution $1/x$, what is the probability density of u?

Solution:

$$P_U(u) = P_X(x) \left| \frac{dx}{du} \right|$$

$$\frac{dx}{du} = 3u^2$$

$$P_U(u) = \frac{1}{x} 3u^2 = \frac{3u^2}{u^3} = \frac{3}{u}$$

This shows the inproper distribution is scale-invariant.

Exercise 23.3

The Dirichlet distribution satisfies a nice additivity property. Imagine that a biased six-sided die has two red faces and four blue faces. The die is rolled N times and two Bayesians examine the outcomes in order to infer the bias of the die and make predictions. One Bayesian has access to the red/blue colour outcomes only, and he infers a two-component probability vector (p_R, p_B). The other Bayesian has access to each full outcome: he can see which of the six faces came up, and he infers a six-component probability vector $(p_1, p_2, p_3, p_4, p_5, p_6)$, where $p_R = p_1 + p_2$ and $p_B = p_3 + p_4 + p_5 + p_6$. Assuming that the second Bayesian assigns a Dirichlet distribution to $(p_1, \ldots, p_6)$ with hyperparameters $(u_1, u_2, u_3, u_4, u_5, u_6)$, show that, in order for the first Bayesian's inferences to be consistent with those of the second Bayesian, the first Bayesian's prior should be a Dirichlet distribution with hyperparameters $\big((u_1 + u_2), (u_3 + u_4 + u_5 + u_6)\big)$.
Hint: a brute-force approach is to compute the integral

$$P(p_R, p_B) = \int d^6\mathbf{p}\; P(\mathbf{p} \mid \mathbf{u})\, \delta\big(p_R - (p_1 + p_2)\big)\, \delta\big(p_B - (p_3 + p_4 + p_5 + p_6)\big).$$

A cheaper approach is to compute the predictive distributions, given arbitrary data $(F_1, F_2, F_3, F_4, F_5, F_6)$, and find the condition for the two predictive distributions to match for all data.

Solution:

Exercise 23.4

N datapoints $\{x_n\}$ are drawn from a gamma distribution $P(x \mid s, c) = \Gamma(x; s, c)$ with unknown parameters s and c. What are the maximum likelihood parameters s and c?

Solution: